

# 100 Days of Linux - Day 097

## Title: Repeating Tasks with while Loops

### Today's Goal

Learn how to use **while** loops to repeatedly execute commands as long as a condition is true.

### Why This Matters

While loops are commonly used for monitoring systems, reading files, waiting for user input and performing continuous automation tasks.

### Commands of the Day

#### Syntax:

```
COUNT=1
while [ $COUNT -le 5 ]
do
    echo $COUNT
    COUNT=$((COUNT+1))
done
```

### Beginner Lesson

A **while** loop keeps running until its condition becomes false. You normally initialise a variable before the loop, update it inside the loop and stop when the condition is no longer true.

### Practice Questions

1. Create a script named `while_loop.sh`.
2. Print the numbers 1 to 5 using a while loop.
3. Display only even numbers from 2 to 10.
4. Print a countdown from 5 to 1.
5. Modify the loop to print your name three times.

### Hints

Use numeric comparison operators such as **-le**, **-lt**, **-ge** and update the counter with **\$((COUNT+1))**.

### Mini Challenge

Write a script that creates five directories named Backup1 through Backup5 using a while loop.

### Common Mistakes

Forgetting to update the counter variable, causing an infinite loop, or using incorrect comparison operators.

### Did You Know?

Many long-running Linux services internally use looping logic to keep checking for new work or events.

### Pro Tip

Always ensure the loop condition eventually becomes false to avoid infinite loops.

### Answer Key (Instructor Only)

Q1: `nano while_loop.sh`

Q2: `COUNT=1; while [ $COUNT -le 5 ]; do echo $COUNT; COUNT=$((COUNT+1)); done`

Q3: Start `COUNT=2` and increment by 2.

Q4: Start `COUNT=5` and decrement using `COUNT=$((COUNT-1))`.

Q5: Loop three times and echo your name.

### Homework

Create a Bash script that prints the multiplication table of 7 using a while loop.

### Next Day Preview

Tomorrow you'll revise everything you've learned about Bash scripting with a comprehensive revision quiz.